

Aurora 2.0 - Progress Summary

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1 Introduction

The AURORA pipeline is designed for long-term excavation monitoring using satellite imagery. The original implementation relied mainly on Sentinel-2 optical imagery and temporal confidence accumulation for excavation detection. Although the system performed well under clear atmospheric conditions, cloud contamination and missing observations created significant uncertainty in temporal monitoring.

The current phase of development focused on two major directions:

- improving the robustness and stability of the original clustering-based excavation pipeline,
- integrating Sentinel-1 Synthetic Aperture Radar (SAR) imagery for cloud-aware gap filling and temporal reconstruction.

The objective of these updates was not to replace the existing Sentinel-2 framework, but rather to improve reliability, reduce false positives, stabilize clustering behavior, and extend monitoring capability into masked or uncertain regions.

2 Pipeline Improvements Prior to SAR Integration

2.1 Dynamic K Experiments and Final Cluster Design

We explored the possibility of dynamically selecting the number of clusters used in the K-Means stage. Two approaches were tested.

The first used silhouette score to estimate the optimal value of K . However, this consistently produced very small cluster counts, which caused spectrally different pixel types to merge into the same cluster and significantly increased false positives.

The second approach attempted to maximize the separation between excavation and non-excavation cluster scores. This also proved to be ineffective

because the excavation score itself is heuristic and does not correspond to a physically meaningful quantity.

These experiments showed that dynamic cluster selection did not provide a meaningful practical advantage. The higher values of K preserved spectral diversity more effectively, while smaller values caused cluster mixing and instability. Based on these observations, the final pipeline adopts a fixed value of:

$$K = 8$$

2.2 Enhanced Cluster Pruning

The cluster pruning stage was strengthened using a statistics-based pruning mechanism.

After identifying the excavation metacluster, clusters are ranked by excavation score. The updated pipeline computes the distance of each candidate cluster from the strongest excavation cluster and applies robust statistical filtering using the median and interquartile range (IQR) of these distances.

Only clusters that are statistically close to the leader cluster and also satisfy the excavation-score threshold are retained. This produces a more conservative and stable pruning mechanism that removes weak excavation-like clusters while preserving true excavation signatures.

2.3 Meta-Clustering Stabilization

The meta-clustering stage was redesigned to reduce the influence of anti-excavation signals such as:

- water bodies,
- shadows,
- dense vegetation.

Previously, these signals were suppressed only after clustering, which still allowed them to distort the distance-based meta-clustering process.

The updated pipeline instead suppresses these signals directly at the feature level before meta-clustering. A copy of the scaled training data is created, and statistical outliers are clamped using median and IQR-based bounds across the full dataset.

This significantly improves the separation between excavation and non-excavation metaclusters because irrelevant spectral extremes no longer dominate the centroid distances.

2.4 Confidence Threshold Evaluation

Dynamic confidence thresholds were also investigated during experimentation. However, the benefits were minimal while the resulting system complexity increased considerably.

The final system therefore retains a fixed temporal confidence threshold of:

60 days

This threshold provided stable behavior across mines and effectively distinguished persistent excavation from temporary seasonal variation.

3 Sentinel-1 SAR Integration

3.1 Motivation

Sentinel-1 SAR imagery provides cloud-independent observations with regular revisit coverage and sensitivity to surface roughness and structural disturbance. Excavated regions often exhibit stronger radar backscatter due to exposed soil, rock surfaces, reduced vegetation cover, and increased surface roughness. These properties make SAR suitable for identifying excavation-related disturbances in regions where Sentinel-2 observations are unavailable or unreliable.

3.2 SAR Gap-Filling Framework

This prevents SAR noise from overriding reliable optical predictions. The SAR integration module operates conservatively. SAR predictions are only used to fill Sentinel-2 unknown or masked pixels; valid Sentinel-2 excavation and non-excavation labels are always preserved. This prevents SAR noise from overriding reliable optical predictions. In practice, Sentinel-2 remains the primary source and SAR acts strictly as a fallback for masked regions.

3.3 SAR Feature Engineering

The SAR implementation currently analyzes:

- VV backscatter,
- VH backscatter,
- VV/VH ratio,
- temporal VV change (ΔVV).

Initial experiments showed that excavation regions generally exhibit:

- higher VV backscatter,
- increased VV relative to vegetation,
- positive temporal disturbance signatures.

4 Adaptive SAR Thresholding

4.1 Problem with Fixed Thresholds

The initial SAR prototype used a global threshold:

$$VV > -12 \text{ dB}$$

Although functional, this proved unstable because SAR response varies across mines, terrain types, moisture conditions, and seasons. A single global threshold therefore produced inconsistent behaviour across locations and required per-site manual tuning.

4.2 Mine-Specific Threshold Learning

To improve robustness, the pipeline now derives SAR thresholds directly from Sentinel-2 confirmed excavation pixels. The process begins by obtaining confirmed excavation pixels from the S2 output, then samples SAR VV values over those regions to estimate a mine-specific distribution. The adaptive threshold is defined as: The implemented threshold estimation is:

$$VV_{threshold} = \mu_{excavated} - \sigma_{excavated}$$

where:

- $\mu_{excavated}$ is the mean VV value over confirmed excavation pixels,
- $\sigma_{excavated}$ is the standard deviation.

4.3 Temporal SAR Disturbance Detection

In addition to absolute VV thresholding, the pipeline introduces disturbance-based SAR analysis.

The system computes:

$$\Delta VV = VV_{current} - VV_{baseline}$$

where:

- $VV_{baseline}$ represents historical SAR behavior,
- $VV_{current}$ represents present SAR response.

A positive delta indicates that the pixel is brighter than its recent seasonal baseline, which is consistent with fresh excavation or sudden surface disturbance. This formulation makes the classifier inherently season-adaptive: as vegetation moisture and surface conditions shift across the year, the baseline shifts with them, so a detection requires genuine deviation from recent norms rather than simply crossing a fixed absolute level.

4.4 Temporal SAR Agreement System

Single SAR observations are highly sensitive to speckle noise, rainfall, and temporary moisture variation. To improve reliability, the pipeline introduces a temporal agreement mechanism. For each masked Sentinel-2 date, nearby Sentinel-1 observations are collected and excavation is accepted only if multiple SAR acquisitions agree. This reduces isolated false positives and transient scattering anomalies that would otherwise inflate candidate excavation counts during wet periods.

4.5 SAR Feature Analysis Framework

A dedicated SAR analysis module was added to study the separability of characteristics.

The system extracts SAR values for:

- confirmed excavation pixels,
- confirmed non-excavated pixels,
- masked regions.

Histogram analysis is generated for:

- VV,
- VH,
- VV/VH ratio.

This enables visual threshold inspection, mine-specific feature evaluation, and provides the empirical foundation for any future classifier development. In particular, the Cohen’s d statistic printed alongside each histogram gives an objective measure of class separability, making it straightforward to determine which feature and which window length best distinguish excavated from non-excavated pixels for a given mine.

5 Output Analysis

The generated excavation progression maps show physically consistent outward expansion, stable temporal growth, and realistic mine evolution patterns. The retroconfirmed excavation area exhibits rapid expansion during early 2022, a long stabilisation period, and renewed expansion during 2024. The candidate excavation curves fluctuate due to temporary detections and SAR noise, while the retroconfirmed curves remain stable due to temporal confidence accumulation. This contrast demonstrates that the confidence system successfully suppresses noisy detections while preserving persistent excavation growth. One of the major risks of SAR integration was excessive false positive generation. The adaptive

threshold system successfully prevented catastrophic SAR over-expansion: excavation growth remained spatially coherent, candidate fluctuations remained bounded, and retroconfirmed excavation remained stable throughout the evaluation period.

6 Conclusion

The updated AURORA 2.1 pipeline successfully extends excavation monitoring beyond purely optical imagery through uncertainty-aware Sentinel-1 SAR integration. The framework now combines improved clustering stability through a fixed $K = 8$, statistics-based cluster pruning, stabilised meta-clustering, adaptive mine-specific SAR thresholding, temporal SAR agreement filtering, and cloud-aware SAR gap filling into a coherent multi-sensor pipeline.

Experimental outputs demonstrate:

- stable excavation progression,
- realistic mine expansion dynamics,
- bounded SAR false positives,
- successful temporal reconstruction in masked regions.

The work represents a transition from single-sensor excavation monitoring toward adaptive multi-sensor temporal reconstruction for long-term mining surveillance.